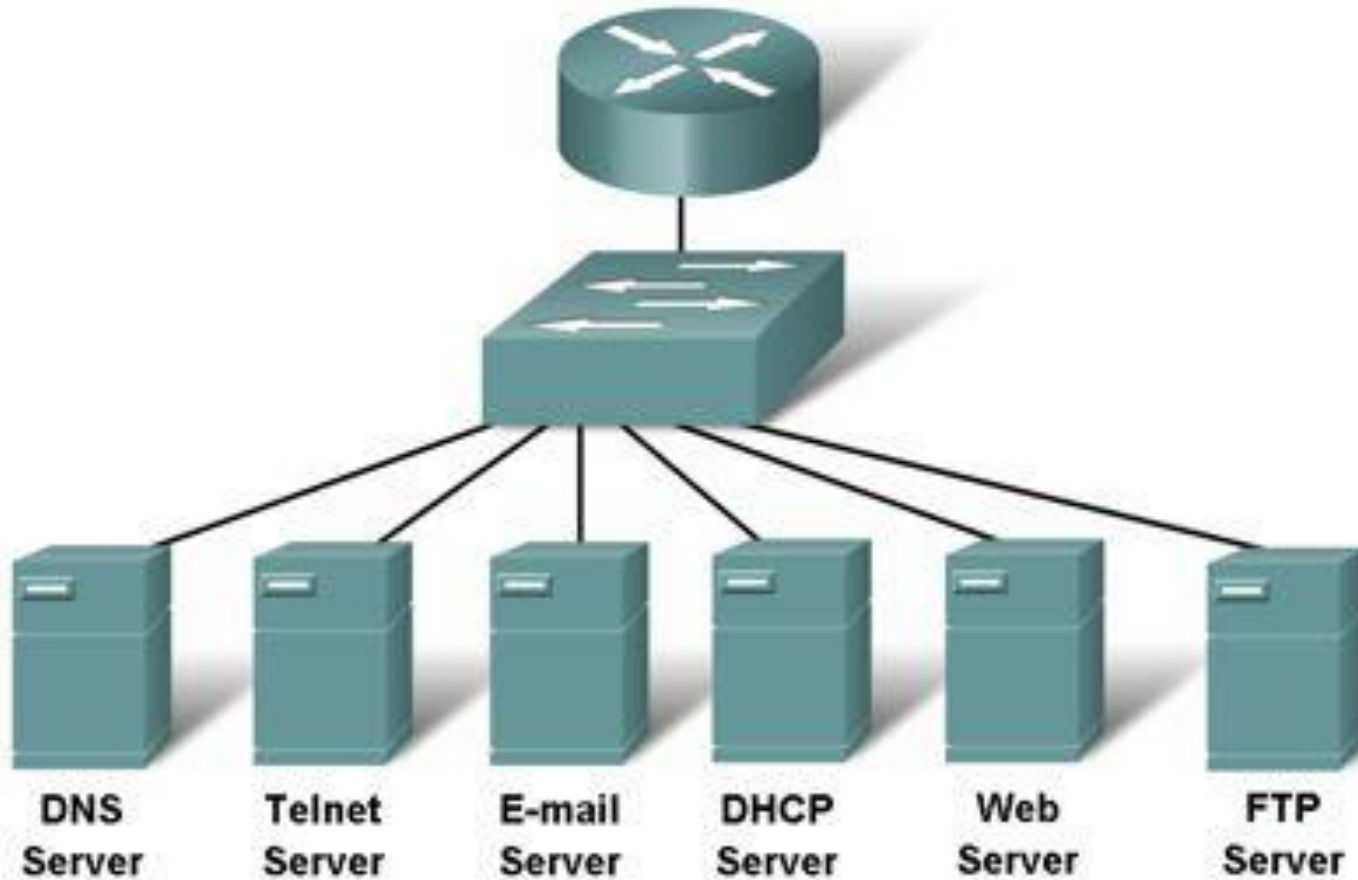


Lecture @ File Transfer Protocol (FTP)

File Transfer Protocol (FTP)

Network Applications



File Transfer Protocol (FTP)

- The developers of early applications conceptually divided methods of network use into two categories: **direct and indirect**
- Direct network applications let a user access a remote host and use it **as if it were local**, creating the illusion that the network doesn't even exist (or at least, minimizing the importance of distance)

File Transfer Protocol (FTP)

- Indirect network use meant getting resources from a remote host and using them on the local system, then transferring them back
- Telnet for direct access and the FTP for indirect network use

TELNET

- TELNET is abbreviation for Terminal Network
- It is standard TCP/IP protocol for virtual terminal services
- TELNET enables establishment of connection to a remote system in such a way that a local terminal appears to be terminal at remote system
- TELNET is general purpose client server application program

File Transfer Protocol (FTP)

- To ensure that files are sent and received **without loss of data** that could corrupt them, **FTP uses the reliable Transmission Control Protocol (TCP)** at the transport layer
- An **authentication** system is used to ensure that only authorized clients are allowed to access a server
- At the same time, a feature sometimes called **anonymous** FTP allows an organization that wishes it to set up a general information server to provide files to anyone who might want to retrieve them

File Transfer Protocol (FTP)

- The interface between an FTP user and the protocol is provided in the form of a set of **interactive user commands**
- After establishing a connection and completing authentication, two basic commands (get, put) can be used to send or receive files
- FTP can also be **used directly by other applications** to move files from one place to another

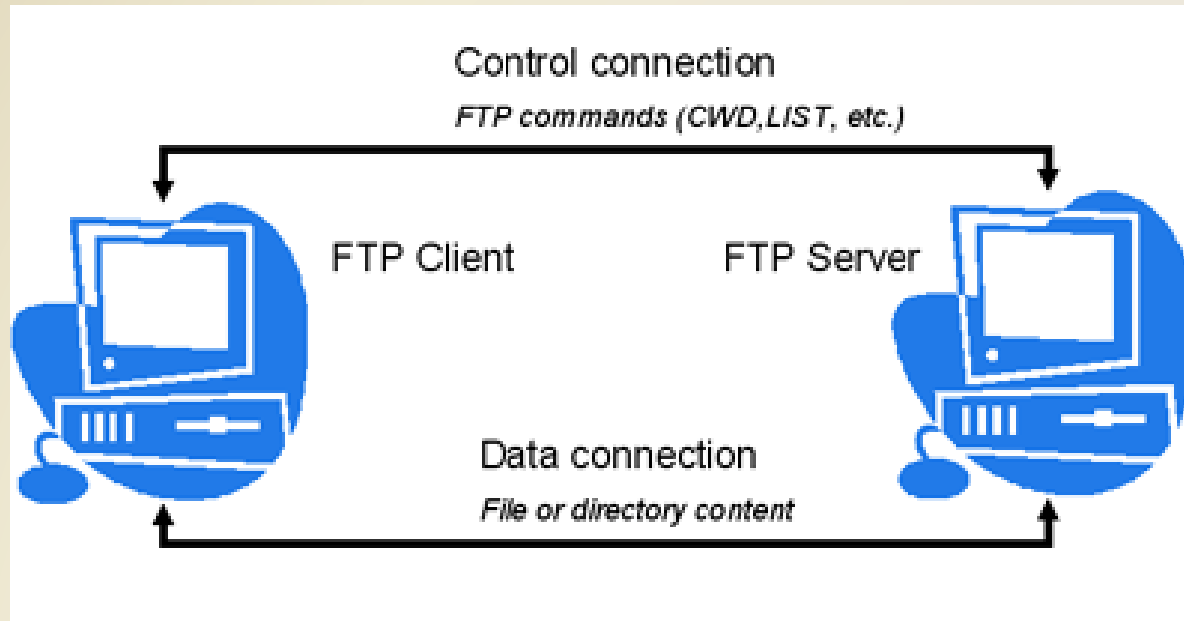
FTP Process Model

- FTP server implementations allow simultaneous access by multiple clients
- Clients use TCP to connect to a server
- The FTP server process awaits connections and creates a slave process to handle each connection
- The slave process accepts and handles a control connection from the client
 - The control connection carries commands that tell the server which file to transfer

FTP Process Model

- An additional TCP connection and process is created to handle each data transfer operation
 - The new TCP connection and process on both the client and server is created for each data transfer operation
- The control connection is kept alive as long as the client keeps the FTP session active
- The data transfer connection is kept alive for the duration of one file transfer
 - For each file that is being transferred, a new data transfer connection is created

FTP Process Model



TCP Port Numbers and Data Connections

- When the client forms an initial connection to a server, the client uses a random protocol port number
- The client contacts the server at a common port number (port 21)
- When a file transfer is initiated, the client obtains an unused port number on its machine

TCP Port Numbers and Data Connections

- The client sends that port number across the control connection to the server
- The client waits for the server to form a TCP connection to that specified port
 - The server uses port 20 for the FTP data transfer